



Jumping genes play a surprising role

A new research indicates that 'jumping genes' also help stabilise 3D folding patterns of DNA molecules. <https://digit.in/feb20-09>



15,000-year old virus found

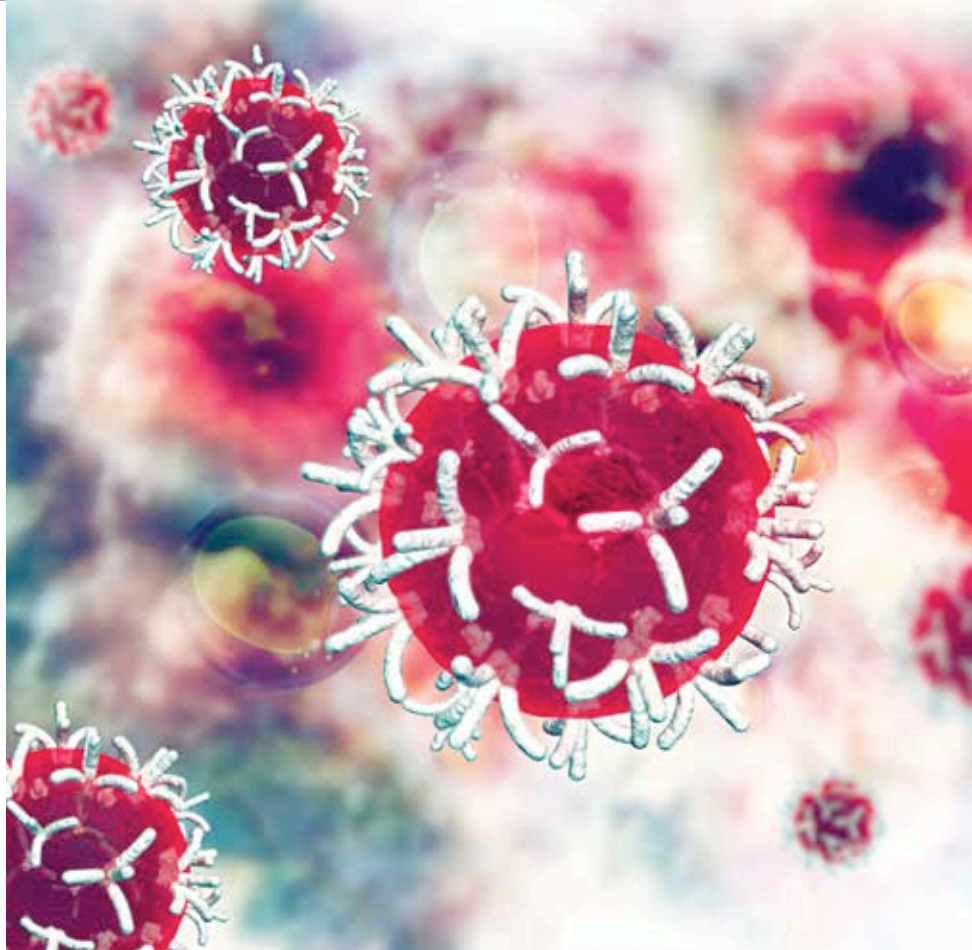
Researchers have discovered an array of ancient virus groups in the ice cores obtained from a Tibetan glacier. <https://digit.in/feb20-10>

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he Cancer Genome Atlas (TCGA) is a landmark project started in 2005 by the US based

National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI). The idea was to make a catalogue of the genetic mutations that cause cancer. This meant collecting tumour samples and blood samples (known as the germ line) from patients, and processing them using gene sequencing and bioinformatics. The TCGA is a continuing effort even after fifteen years, and has generated over 2.5 petabytes of data for over 11,000 patients. This data is available to researchers all around the world, and has been used to develop new approaches to diagnose, treat and prevent cancer.

The data collection is done with the consent of the patients, who are informed that such data will benefit those who are affected or could be affected by cancer in the future. The data collection has also been designed in a way to make sure that the privacy of the patients is maintained, even though researchers have access to the data. Data of over 33 types of cancer are collected by TCGA. The researchers using the database have also developed a number of computational tools to visualise and process the database. These tools allow researchers to not only explore particular kinds of cancers across different patients, but also compare different kinds of cancers. Dr Anand Deshpande, Founder, Chairman and Managing Director of Persistent Systems explains the benefit of the TCGA, "there is a belief that if we are able to compare and look at data, at different individuals and different cancers, then our ability to understand cancer would be a lot better. One of overall long term goals in cancer research is to see if you can create a personalised, targeted drug or therapy. There have been some that have gotten created in the past, where one single shot basically fixes cancer for good.



A LOCAL CANCER GENOME ATLAS

The Cancer Genome Atlas has proved to be a valuable platform for research all around the world, and now there is an effort to initiate such a project in India.

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This is possible in certain cases if you have a very clear understanding of exactly what that particular cancer is likely to do to that individual. No two cancers are alike actually. Every cancer seems to be unique, it has its

own unique manifestation. Getting it right at that level can provide tremendous improvement in terms of getting targeted therapy, where the cancer is cured and all the side effects are managed. Basically, you can cure cancer, it is just a disease that gets cured, the patient does not need therapy or go into remission or any of those things.